VITyarthi | Open Source Software (CSE0002)

The Open Source Audit

A Capstone Project for OSS NGMC Course

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Introduction :

Linux and Free, Open Source Software (FOSS) are part of modern computing. Open-source software provides the opportunity for developers to access, modify, and freely share source code which helps to encourage innovation and teamwork among developers. Some of the most common software that represents FOSS includes Python, a high level programming language famous for code clarity and readability. Web development, data science, AI and Automation can use Python. Python is open-source which means its source code is publicly available and can be modified by anybody. Wide and global supportive community, mainly used for scripting and development in Linux systems. This project aims to provide information relating major concepts of Linux and FOSS (Free & Open Source Software) in open-source software on Python! Including the Hands-On implementation using shell script tasks on Linux.

Origin And Philosophy – (Part-A) :

❖ A.1 The Problem This Software Was Created to Solve

Prior to the release of open-source software, proprietary systems largely ruled the computing industry. These systems didn't allow users to see or modify the source code and theoretically made it so that people either couldn't understand how an application worked, or they couldn't customize it at all.

Therefore, depending on the software companies for any updates, bugs fixes, or feature additions were heavily dependent. This dependency had the dual impact of increasing costs and stagnating innovation, as users could not be involved in polishing the software. Programming languages like C and C++ ruled the field, but they needed a deep knowledge of complex syntax and memory allocation. This made them less beginner-friendly and it negatively affected overall productivity. It needed an easy language, easy to learn and powerful programming language.

Python is an open-source programming language that was developed to solve these problems, emphasizing simplicity and readability. It provided developers with a very concise and easy-to-read syntax to avoid complexity from the code. Meanwhile, Linux offered a free and open operating system that allowed complete control over the computing environment. Linux and

Python solved a lot of pain points in system-level and programming-level computing, all under the themes of accessibility, flexibility, and efficiency. Meanwhile, Linux offered a free operating system that users could run whatever software they pleased on. Linux and Python provided solutions to system-level and programming-level problems encapsulating openness, flexibility, and efficiency.

❖ A2. The License — What It Actually Says

Licensing is a fundamental aspect of open-source software, as it defines how software can be used, modified, and distributed. The Linux operating system is released under the GNU General Public License, which ensures that the software remains free and open for all users. This license allows users to access the source code, make modifications, and distribute the software, provided that any modified version is also shared under the same license.

Python, on the other hand, is distributed under the Python Software Foundation License, which is a more permissive open-source license. It allows users to freely use, modify, and distribute Python, even for commercial purposes, without strict restrictions. The concept of copyleft, which is a key feature of the GPL, ensures that software remains open and prevents it from being converted into proprietary software. These licensing models play a crucial role in maintaining the openness and fairness of software development. They encourage collaboration and ensure that the benefits of software are shared among all users.

❖ A3. The Ethics of Open Source

The open-source movement is not only based on technical principles but also on strong ethical values. It promotes the idea that knowledge should be freely available and that collaboration leads to better results. Open-source software encourages developers to share their work, learn from others, and contribute to a global community.

The core ethical principles of open source include freedom, transparency, collaboration, and accessibility. Freedom ensures that users have control over the software they use. Transparency allows users to examine the source code, which increases trust and security.

Collaboration enables developers from different parts of the world to work together, leading to rapid innovation. Accessibility ensures that software is available to everyone, regardless of their financial background. Organizations such as the Free Software Foundation and the Open Source

Initiative play a significant role in promoting these ethical values. Python is a perfect example of these principles in action, as it is developed and maintained by a global community that continuously works to improve it.

Linux Footprint – (Part-B) :

Linux has established itself as one of the most powerful and widely used operating systems in the modern technological world. Since its development, it has grown from a simple open-source project into a global platform that powers a vast range of devices and systems. The Linux kernel, developed by Linus Torvalds, forms the core of this ecosystem and provides a stable and secure foundation for various operating system distributions. Its open-source nature allows developers to modify and customize it according to their needs, which has contributed significantly to its widespread adoption.System Analysis and Observations:

- Python was installed using the apt package manager, a standard tool for managing software in Debian-based systems
- The installation was verified using the command python3 --version
- Python executables are located in system directories such as /usr/bin
- The system runs on Ubuntu 24.04 LTS with a WSL2 kernel

The following screenshots confirm Python installation and configuration on the Linux system:

```
shalini_anand813@hp:~$ sudo apt install python3
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.12.3-0ubuntu2.1).
python3 set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 63 not upgraded.
shalini_anand813@hp:~$
```

The FOSS Ecosystem – (Part-C) :

Python is an important part of the Free and Open Source Software (FOSS) ecosystem, which consists of a large network of developers, tools, libraries e.t.c.

Key Characteristics of the FOSS Ecosystem:

- Large and active global developer community
- Availability of thousands of open-source libraries
- Continuous updates and improvements
- Strong documentation and community support

Major Application Areas:

- **Web Development:** Frameworks
- **Data Science:** Libraries such as NumPy and Pandas
- **Artificial Intelligence:** Machine learning and deep learning applications
- **Automation:** Task automation

The FOSS ecosystem encourages innovation and collaboration, ensuring that software evolves continuously to meet modern technological demands.

Open Source Vs Proprietary – (Part-D) :

Feature	Open Source (Python)	Proprietary Software
Source Code	Available	Not Available
Cost	Free	Paid
Customization	High	Medium
Transparency	High	Low
Community Support	Strong	Medium

Open-source software provides flexibility, transparency, and cost-effectiveness. Users can modify and improve the software according to their needs.

In contrast, proprietary software offer controlled environments, better official support, and enhanced security in some cases, but limits user freedom.

For learning, experimentation, and innovation, open-source software is more advantageous.

Shell Scripts :

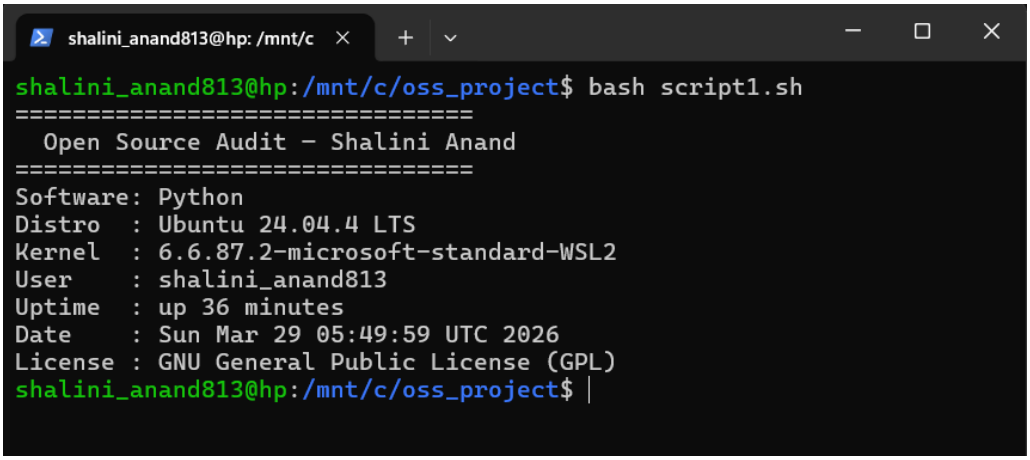
Shell scripting plays a crucial role in automating tasks and managing system operations in Linux environments. In this project, five scripts were used to show different functions.

❖ *Script 1 – System Identity Report*

This script collects and displays key system information, including:

- Username
- Kernel version
- System uptime
- Linux distribution
- Current date

It provides a quick overview of the system and environment :

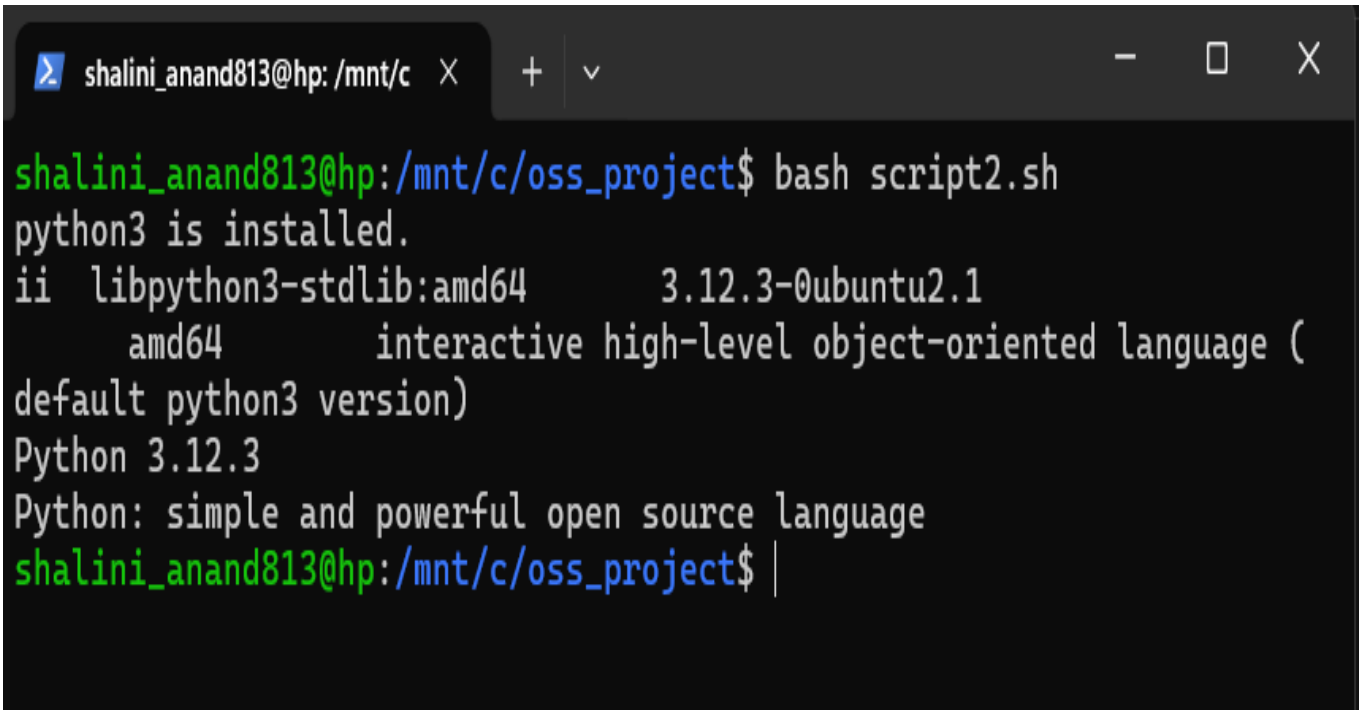


❖ *Script 2 – Package Inspector*

This script checks whether Python is installed on the system. It:

- Verifies installation status
- Displays package details
- Shows Python version

Conditional statements are used to provide appropriate output based on system status.



❖ *Script 3 – Disk and Permission Auditor*

This script analyzes important system directories and provides:

- File and directory permissions
- Ownership details
- Disk usage information.

It helps in understanding system structure and access control mechanisms.

```
shalini_anand813@hp: /mnt/c
shalini_anand813@hp:/mnt/c/oss_project$ bash script3.sh
Directory Audit Report
----------------------
/etc => Permissions: drwxr-xr-x root root | Size: 4.3M
/var/log => Permissions: drwxrwxr-x root syslog | Size: 33M
/home => Permissions: drwxr-xr-x root root | Size: 44K
/usr/bin => Permissions: drwxr-xr-x root root | Size: 95M
/tmp => Permissions: drwxrwxrwt root root | Size: 28K

Python Config Directory:
drwxr-xr-x 2 root root 4096 Feb 10 00:55 /etc/python3
shalini_anand813@hp:/mnt/c/oss_project$
```

❖ *Script 4 – Log File Analyzer*

This script scans system log files for a specific keyword such as “error”. It:

- Counts the number of occurrences
- Displays recent matching entries

```
shalini_anand813@hp: /mnt/c
shalini_anand813@hp:/mnt/c/oss_project$ bash script4.sh /var/log/dpk
g.log error

Keyword 'error' found 39 times in /var/log/dpkg.log

Last 5 matching lines:
2026-02-10 00:56:02 status unpacked liberror-perl:all 0.17029-2
2026-02-10 00:56:32 configure liberror-perl:all 0.17029-2 <none>
2026-02-10 00:56:32 status unpacked liberror-perl:all 0.17029-2
2026-02-10 00:56:32 status half-configured liberror-perl:all 0.17029
-2
2026-02-10 00:56:32 status installed liberror-perl:all 0.17029-2
shalini_anand813@hp:/mnt/c/oss_project$
shalini_anand813@hp:/mnt/c/oss_project$
```

Observation:

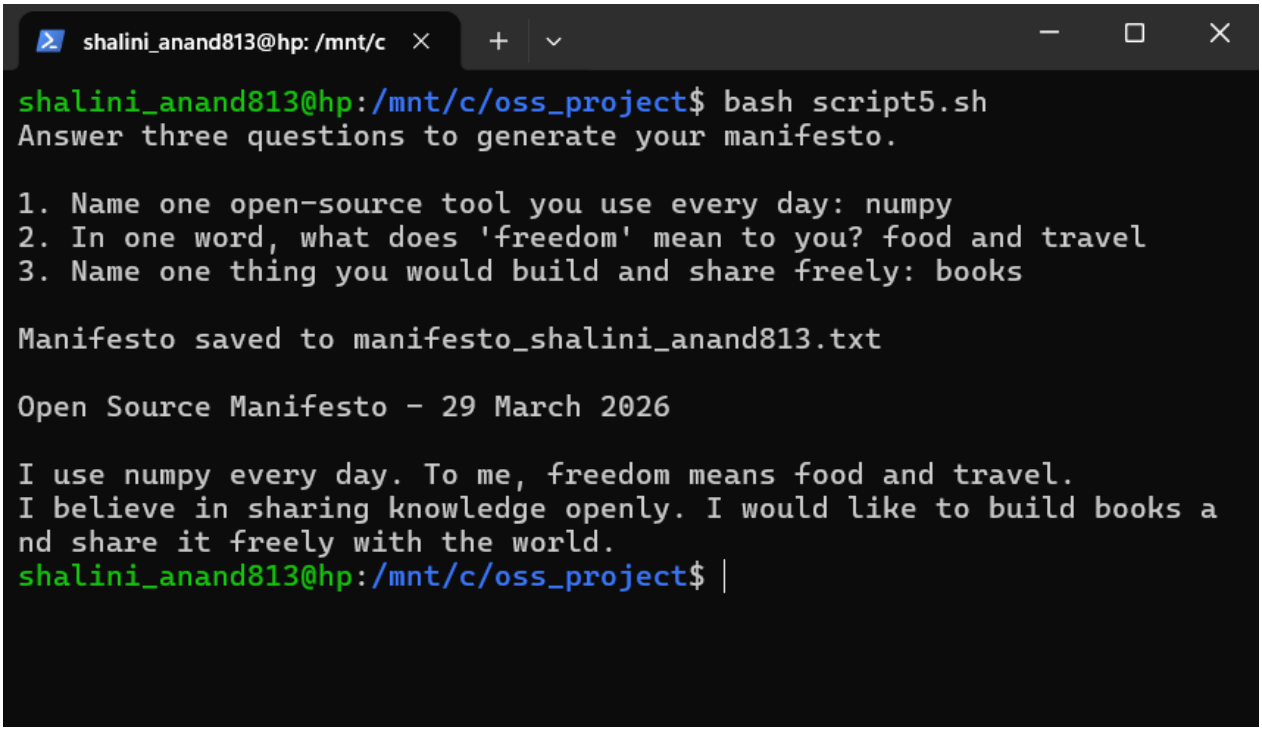
Some matches may not represent actual system errors, as the keyword can appear within package names. This highlights a limitation of simple keyword-based analysis and suggests the need for more advanced filtering techniques.

❖ *Script 5 – Manifesto Generator*

This script interacts with the user by taking input and generating a personalized open-source manifesto. It:

- Accepts user responses
- Creates a text file
- Displays the generated content

It Shows user interaction, file handling, and output formatting in shell scripting.



Project Timeline :

Day	Task
Day1	*Setup WSL and install Ubuntu*
Day2	*Install Python and explore Linux environment*
Day3	*Develop and test Script 1 and Script 2*
Day4	*Develop Script 3, Script 4, and Script 5*
Day5	*Perform testing, capture screenshots, and prepare report*

REFERENCES :

- **Torvalds, L. (1991). Linux Kernel Development.** : https://www.kernel.org
- **Van Rossum, G. Python Programming Language**. : https://www.python.org
- **Free Software Foundation.** : https://www.gnu.org/philosophy/free-sw.html
- **Open Source Initiative**. : https://opensource.org/osd
- **Python Software Foundation**. : https://docs.python.org/3/license.html
- **GitHub. About Open Source and Collaboration**. : https://docs.github.com
- **The Linux Documentation Project. Linux System Documentation**. : https://tldp.org
- **GeeksforGeeks. Linux and Python Tutorials**. : https://www.geeksforgeeks.org

CONCLUSION :

This project provides a hands – on experience in open-source software, Linux systems, and shell scripting. It shows how Python can be effectively used within an open-source environment and how Linux tools can be used for system monitoring and automation.
The use of WSL simplified the process by providing linux in windows system. Through this project, key concepts such as open-source Definitions, licensing, system operations, and scripting were understood both by theory and practically.
Overall, the project enhanced technical skills, problem-solving ability, and understanding of modern software development practices, making it a valuable learning experience.